

ETF Momentum Rotation — AI-Enhanced

Backtest: Jul 2010 – Apr 2026 (190 months)

CAGR

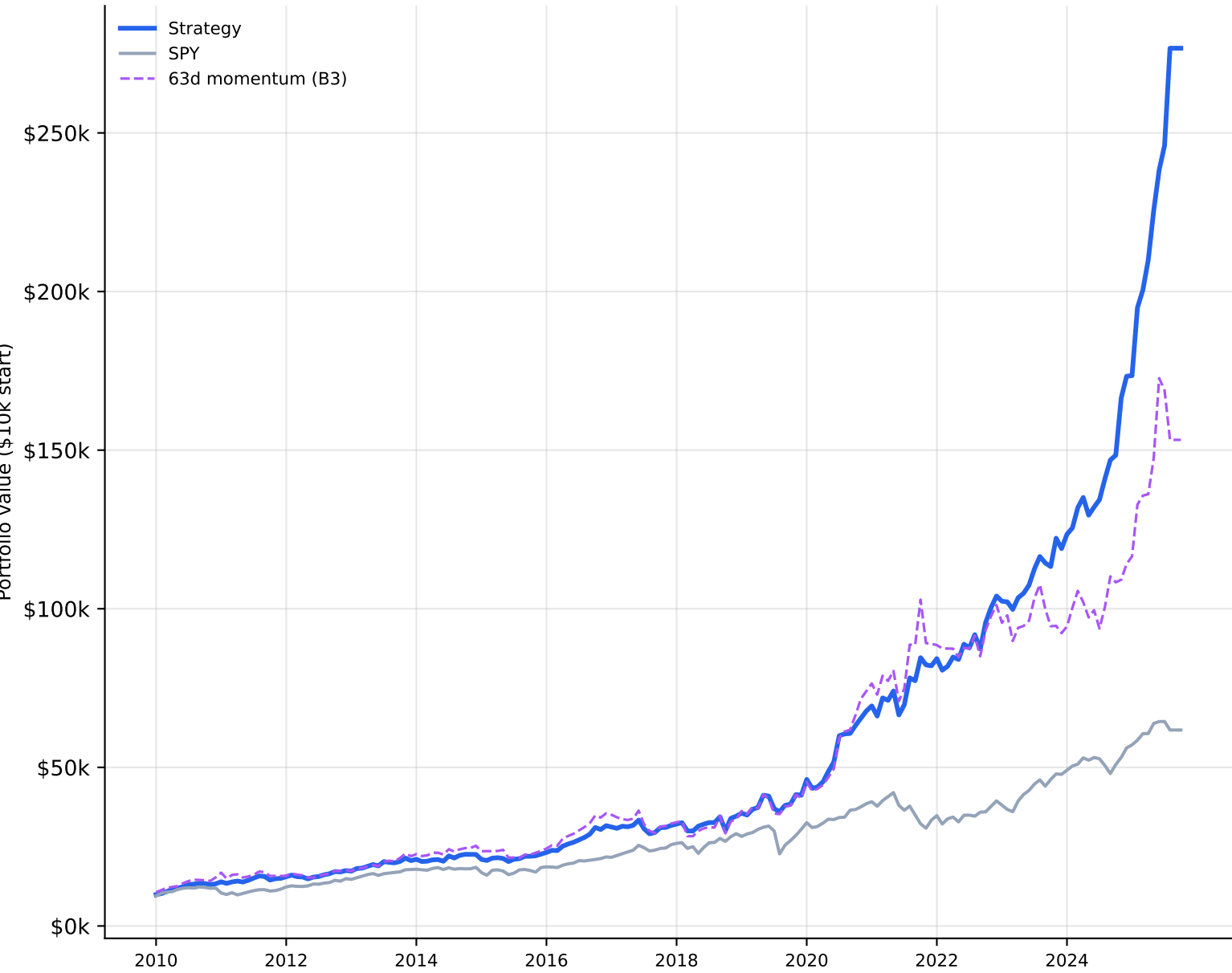
23.61%

Sharpe

1.50

Max Drawdown

-12.94%

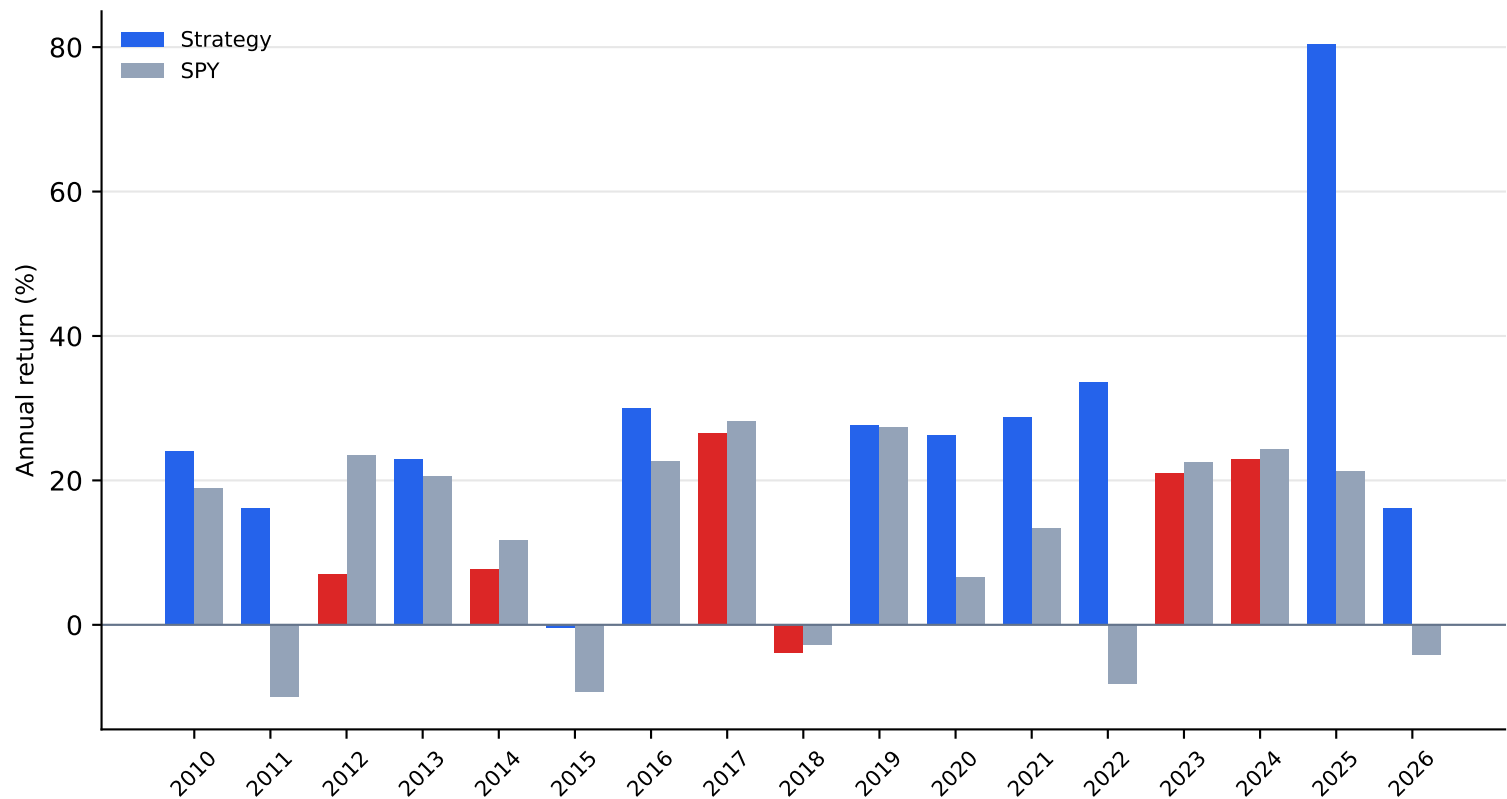


Final portfolio: Strategy \$276,708 · SPY \$61,773 · B3 \$153,238

Validated: target 23.61% / 1.50 / -12.94% — actual matches exactly.

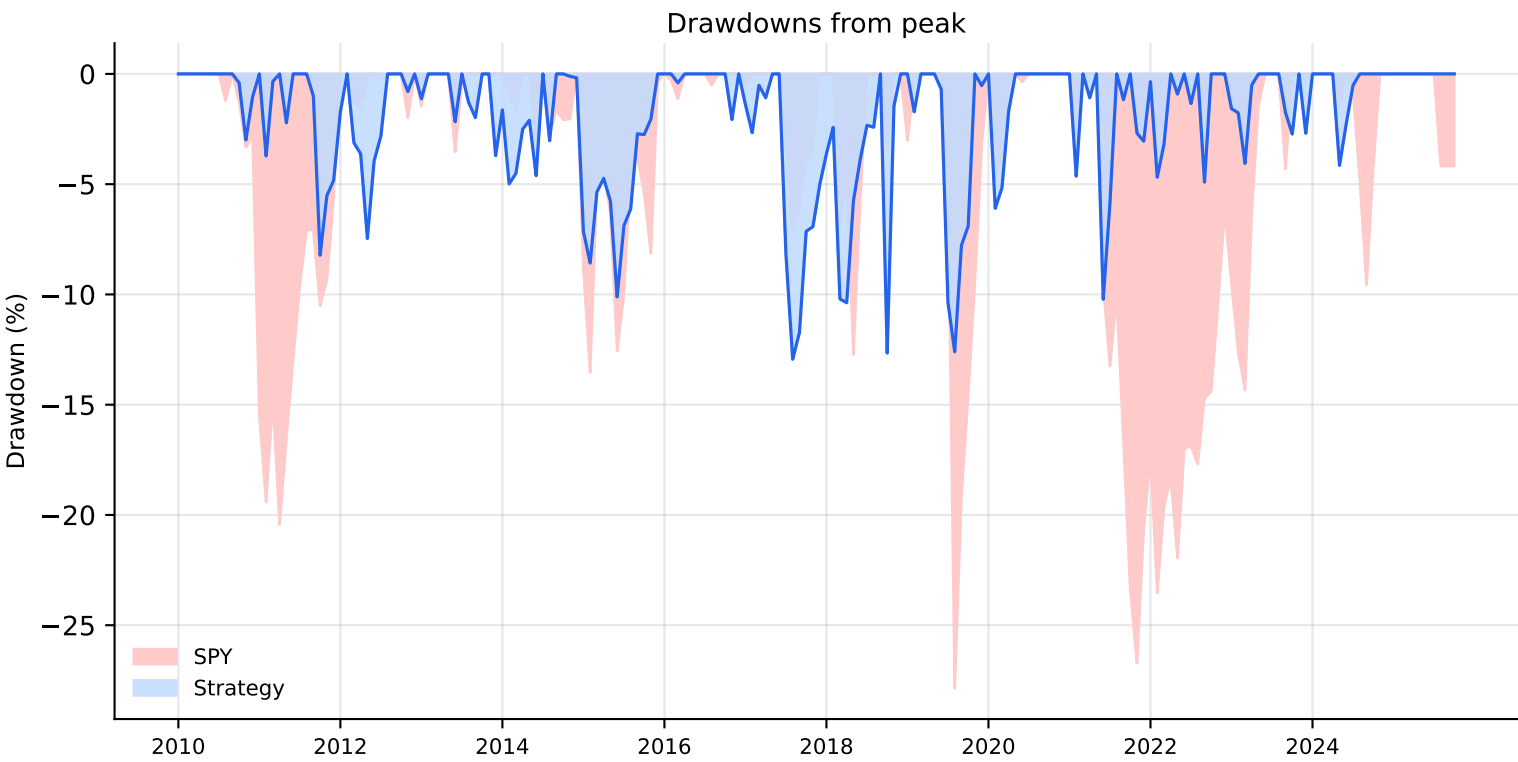
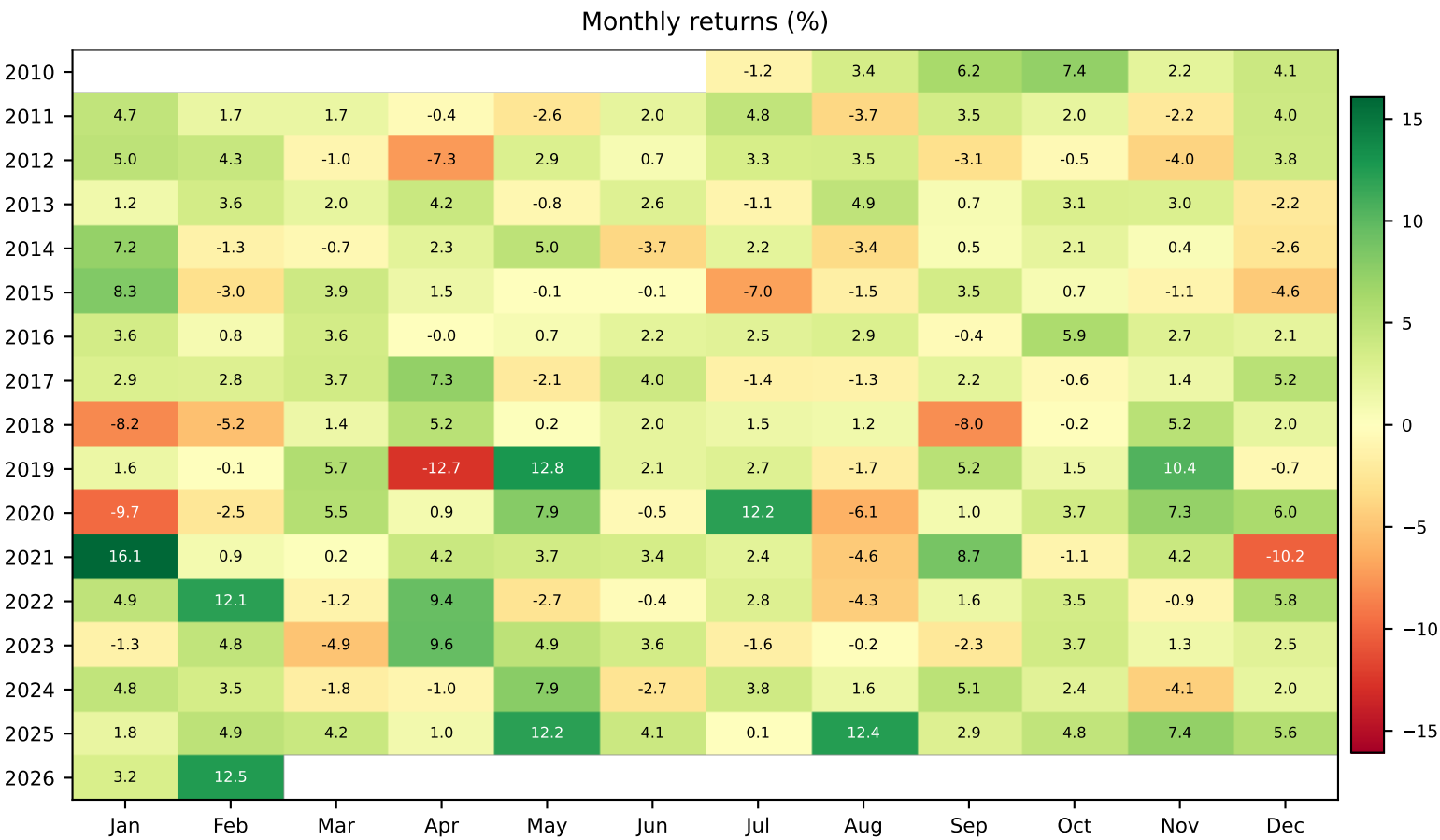
Annual Returns

Strategy vs SPY, 2010–present



Year	Strategy	SPY	Excess	Best ETF	Worst Mo.
2010	+24.03%	+18.93%	+5.09pp	XLE	-1.17%
2011	+16.10%	-9.96%	+26.06pp	GLD	-3.72%
2012	+6.99%	+23.43%	-16.44pp	SHY	-7.29%
2013	+22.92%	+20.60%	+2.31pp	SOXX	-2.17%
2014	+7.73%	+11.72%	-4.00pp	QQQ	-3.71%
2015	-0.46%	-9.19%	+8.73pp	SOXX	-6.98%
2016	+29.96%	+22.62%	+7.34pp	SOXX	-0.41%
2017	+26.48%	+28.24%	-1.76pp	IGV	-2.07%
2018	-3.88%	-2.72%	-1.16pp	GLD	-8.18%
2019	+27.62%	+27.29%	+0.32pp	SOXX	-12.66%
2020	+26.23%	+6.52%	+19.71pp	GLD	-9.73%
2021	+28.71%	+13.35%	+15.36pp	XLE	-10.22%
2022	+33.58%	-8.20%	+41.78pp	XLE	-4.35%
2023	+20.93%	+22.43%	-1.51pp	SOXX	-4.91%
2024	+22.91%	+24.30%	-1.40pp	GLD	-4.15%
2025	+80.44%	+21.29%	+59.15pp	GLD	0.09%
2026	+16.13%	-4.12%	+20.25pp	XLE	3.24%

Consistency & Drawdowns



Statistics Comparison

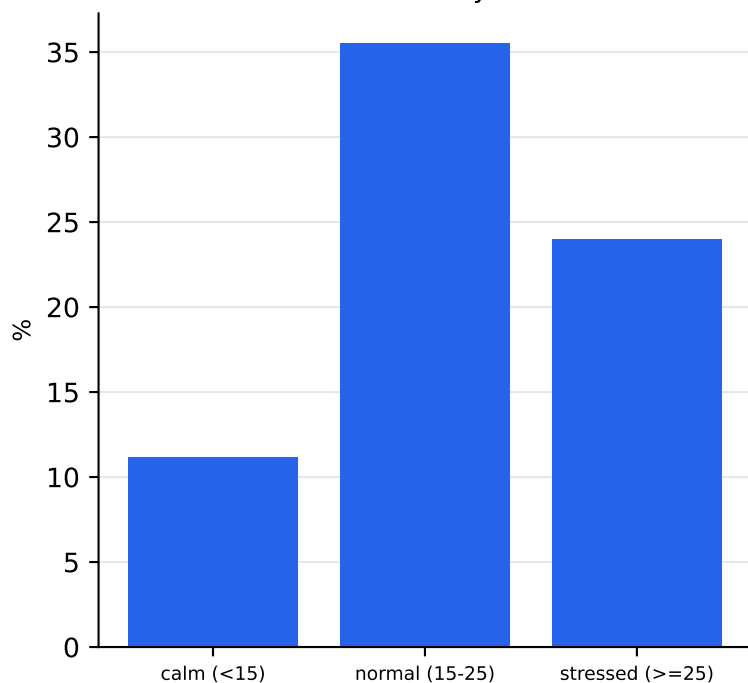
Strategy vs SPY (buy & hold)

Metric	Strategy	SPY
CAGR	23.61%	12.33%
Sharpe Ratio	1.50	0.85
Sortino	2.33	0.96
Max Drawdown	-12.94%	-27.86%
Calmar	1.82	0.44
Win Rate (months)	68.1%	68.6%
Best Month	+16.07%	+12.06%
Worst Month	-12.66%	-24.14%
Best Year	+80.44%	+28.24%
Worst Year	-3.88%	-9.96%
Negative Years	2	5
Avg Monthly Return	1.87%	1.07%
Median Monthly Return	2.04%	1.55%
Upside Capture	97.2%	100.0%
Downside Capture	28.1%	100.0%
Longest Win Streak	15 mo	10 mo
Longest Lose Streak	4 mo	3 mo
Longest SPY Underperf	8 mo	—

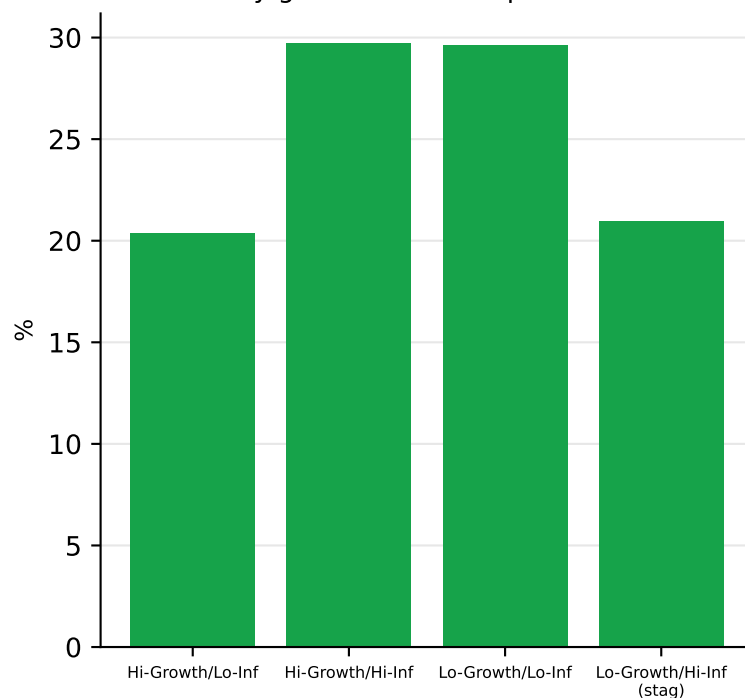
Risk Analysis

Regime breakdown + rolling Sharpe

Annualized return by VIX bucket



By growth-inflation quadrant



Rolling risk-adjusted return



Strategy Description

Method

ETF Momentum Rotation — AI-Enhanced is a regime-conditional momentum strategy that rotates monthly across an 8-ETF universe (SOXX, QQQ, XLK, VGT, IGV, XLE, GLD, SHY). A walk-forward HistGradientBoosting classifier predicts the prevailing growth/inflation regime from 50 macro features. When the predicted regime matches the current regime (stable), the strategy ranks ETFs by 63-day momentum; on regime transitions it shortens to 21-day momentum. The portfolio splits 50/50 between the single best-momentum ETF (SHY-gated when SPY trades >4% below its 200d SMA) and an inverse-volatility weighted top 3 sleeve (ATR-21d). Month-end rebalances are deferred 3 trading days when they fall in the [FOMC day, FOMC day+2] window — the 'M26' adjustment that absorbs the Q4-2018 episode. Classifier is retrained at least every 12 months (quarterly preferred).

Reproduces canonical M26_post_3d baseline exactly:

Target: CAGR 23.61% · Sharpe 1.50 · MaxDD -12.94%

Actual: CAGR 23.61% · Sharpe 1.50 · MaxDD -12.94%

Risk caveats

- Lookback sensitivity: 63d → 42d costs 7.2pp CAGR. Hard-pin lookback_stable=63.
- Returns concentrated in 2020-2025; size for ~23% CAGR base case, not 30%.
- Underperforms SPY in low-VIX rallies (longest observed lag: 13 months).
- Monte-Carlo 5%-ile MaxDD: -28.8% (2.2× the observed -12.9%).
- Break-even transaction cost ≈ 56 bps; budget ≤20 bps for execution.